Signed Additive Forgetting Attention for Generative Recommendation

Abstract

Recommendation histories mix durable preferences with transient intent. HSTU represents these histories with signed, unnormalized pair coefficients, so multiple relevant actions can accumulate rather than compete for unit probability mass. However, an HSTU layer has no explicit path variable that marks whether an old action and the current query are separated by a change of intent. We introduce *Signed Additive Forgetting Attention* (SAFA), which assigns each event a per-head retention gate and multiplies every HSTU pair by the survival product along its intervening path. This preserves HSTU's signed SiLU kernel, relative time and position bias, and additive value aggregation, while making intent boundaries explicit. Across controlled three-seed comparisons on MovieLens-1M and MovieLens-20M, SAFA improves mean NDCG@10 from 0.19016 to 0.19575 and from 0.21094 to 0.21512, respectively; all six paired deltas are positive. These results support path-dependent forgetting as a useful inductive bias for generative recommendation.

Keywords: generative recommendation, HSTU, signed attention, forgetting, sequential recommendation

1. Introduction

Users do not express one stationary preference. A history may contain a durable interest in science fiction, a short search for children's books, and then a return to the original interest. A sequential recommender must accumulate repeated evidence within an intent while preventing an intervening change of intent from making stale evidence dominate the next prediction.

- **HSTU accumulates signed evidence.** The Hierarchical Sequential Transduction Unit (HSTU) [Zhai et al., 2024] assigns, for head h, the causal pair coefficient

$$a_{ij}^{\mathrm{H},h} = \frac{M_{ij}}{N}\,\mathrm{SiLU}\Big((\boldsymbol{q}_i^h)^\top \boldsymbol{k}_j^h + b_{ij}\Big), \tag{1}$$

  where b_{ij} contains learned relative position and time bias. Unlike softmax attention, these coefficients are signed and not row-normalized. Compatible actions therefore reinforce one another, while a negative coefficient can subtract an anti-aligned value direction.

- **The missing operation is an explicit intent boundary.** At a fixed layer, Equation (1) forms a direct pair from its projected endpoints and their relative bias. Contextual representations can encode the intervening history implicitly, but there is no compositional path factor requiring one boundary event t to attenuate every pair $j < t \le i$ that crosses it. Each stale endpoint must instead be suppressed separately.

- **SAFA adds path-dependent survival.** Each event emits a per-head retention $f_t^h \in (0,1)$. Its product along the path from key j to query i defines

$$F_{ij}^h = \prod_{t=j+1}^{i} f_t^h, \qquad a_{ij}^{\mathrm{S},h} = F_{ij}^h a_{ij}^{\mathrm{H},h}. \tag{2}$$

If f_t^h is small, every historical pair spanning t is attenuated coherently. Evidence still adds within a stable intent, but decays across a learned boundary. Because F_{ij}^h is nonnegative and introduces no row normalization, this operation preserves both the sign of HSTU coefficients and their additive aggregation.

Our contribution is a minimal extension of HSTU rather than a new sequence block. We (i) introduce key-conditioned path survival while retaining HSTU's pair kernel, relative bias, and output transformation; (ii) isolate this change in a paired HSTU control; and (iii) show consistent gains over six paired runs across MovieLens-1M and MovieLens-20M. The method remains explicit-pair quadratic attention; "additive" describes how evidence is combined, not its asymptotic complexity.

2. Signed Additive Forgetting Attention

Motivation. An HSTU layer combines pointwise projection, spatial aggregation, and pointwise transformation [Zhai et al., 2024]. Given $\boldsymbol{x}_i \in \mathbb{R}^d$, the projection produces gate, value, query, and key features $\boldsymbol{u}_i, \boldsymbol{v}_i, \boldsymbol{q}_i, \boldsymbol{k}_i$. Head h then assigns each past event j the signed coefficient in Equation (1), aggregates its value, and passes the concatenated heads through normalization, gating, projection, and a residual connection.

This pair coefficient exposes the missing operation. At a fixed layer, the score for (i, j) depends on its projected endpoints and their relative bias, but not on an explicit path variable between them. If an intervening event t marks a change of intent, HSTU may learn to suppress one stale pair, yet it has no mechanism by which the same event must attenuate every pair $j < t \leq i$ that crosses that boundary. The effect must be inferred separately for each older key. We therefore seek a minimal extension in which boundaries compose along the sequence while HSTU's signed, unnormalized accumulation remains unchanged.

Architecture. SAFA equips each position and head with a key-conditioned retention,

$$f_t^h = \sigma((\boldsymbol{w}_f^h)^\top \boldsymbol{k}_t^h + c_h), \tag{3}$$

and defines the survival of event j at query i by

$$F_{ij}^h = \prod_{t=j+1}^{i} f_t^h, \qquad F_{ii}^h = 1, \qquad a_{ij}^{\mathrm{S},h} = \frac{M_{ij}}{N} F_{ij}^h \, \mathrm{SiLU}(s_{ij}^h), \tag{4}$$

where M_{ij} is the causal validity mask and N is the padded sequence length. A small f_t^h coherently attenuates all older evidence whose path crosses t, and products naturally compose multiple boundaries. Because the nonnegative survival factor multiplies the HSTU coefficient outside SiLU, it changes magnitude without reversing sign or renormalizing surviving events. SAFA then uses the original HSTU value aggregation, pointwise transformation, and residual update. Table 1 summarizes the complete layer and highlights the only added computation.

Operation	HSTU	SAFA
Pointwise projection	$\bar{x}_i = \mathrm{LN}(x_i), \quad [u_i, v_i, q_i, k_i] = \mathrm{Split}(\mathrm{SiLU}(W_1\bar{x}_i))$	
Pair score	$s_{ij}^h = (q_i^h)^\top k_j^h + b_{ij}$	
Retention gate	$f_t^h \equiv 1$	$f_t^h = \sigma((w_f^h)^\top k_t^h + c_h)$
Path survival	$F_{ij}^h \equiv 1$	$F_{ij}^h = \prod_{t=j+1}^{i} f_t^h, \quad F_{ii}^h = 1$
Pair coefficient	$a_{ij}^{\mathrm{H},h} = \frac{M_{ij}}{N}\mathrm{SiLU}(s_{ij}^h)$	$a_{ij}^{\mathrm{S},h} = \frac{M_{ij}}{N} F_{ij}^h \mathrm{SiLU}(s_{ij}^h)$
Spatial aggregation	$z_i^\star = \mathrm{Concat}_h\left(\sum_{j\le i} a_{ij}^{\star,h} v_j^h\right), \quad \star \in \{\mathrm{H}, \mathrm{S}\}$	
Pointwise transformation	$x_i^+ = x_i + W_o(u_i \odot \mathrm{LN}(z_i))$	
Parameters (ML-1M)	313,000	313,416 (+416, +0.13%)

Table 1: HSTU and SAFA layers. Blue marks the only new computation and its insertion; all projections, pair scores, relative time and position biases, value aggregation, normalization, and residual output are inherited unchanged from HSTU. Counts use the MovieLens-1M architecture; dormant control tensors are not counted as HSTU parameters.

Initialization. We initialize $w_f^h = 0$ and assign heads different initial horizons. For a time constant T_h, setting

$$\bar{f}_h = e^{-1/T_h}, \qquad c_h = \mathrm{logit}(\bar{f}_h) \tag{5}$$

gives $F_{ij}^h = e^{-(i-j)/T_h}$ at initialization; we log-space T_h from 8 to 256 and then learn content-dependent gates. A layer adds only $H(d_k + 1)$ parameters.

Efficiency and hardware optimization. SAFA preserves HSTU's dominant $O(N^2 d)$ attention complexity. It adds an $O(NHd_k)$ gate projection, an $O(NH)$ prefix scan, and one exponential and multiplication per valid pair. A naive implementation materializes the survival matrix $F \in \mathbb{R}^{B\times H\times N\times N}$ and incurs an additional quadratic global-memory pass. We develop an optimized Triton implementation that computes the segmented log-retention prefix $P_i^h = \sum_{t\le i} \log f_t^h$ and evaluates $F_{ij}^h = \exp(P_i^h - P_j^h)$ within each tiled HSTU attention block. Fusing survival with the signed pair coefficient reduces auxiliary survival storage from $O(BHN^2)$ to $O(BHN)$ while preserving causal masking, relative bias, SiLU, and unnormalized value aggregation. We validate the fused forward and backward operators against the dense PyTorch reference.

3. Experiments

Setup. We evaluate the academic retrieval setting from the HSTU release on MovieLens-1M, MovieLens-20M [Harper and Konstan, 2015], and Amazon Reviews (Books). The ML-1M model uses 8 blocks, 2 heads, and $d_q = d_k = d_v = 25$; ML-20M uses 16 blocks, 8 heads, and $d_q = d_k = d_v = 32$. Both use maximum sequence length 200, sampled-softmax loss with 128 negatives, and 101 epochs. Amazon Books uses maximum sequence length 50, 16 blocks, 8 heads, $d_k = d_v = 8$, 512 negatives, and 201 epochs. We pair seeds 42–44 and keep the data order, initialization, optimizer, and all non-attention configuration fixed. We denote SAFA on this backbone as SAFA-large. On ML-1M, SAFA-large adds 416 parameters to HSTU-large's 313,000 (+0.13%); on ML-20M, it adds 4,224 to 38,913,120 (+0.011%); and on Amazon Books, it adds 1,152 to 44,865,440 (+0.0026%). For matched training plumbing, the HSTU control instantiates the same gate tensors but removes them from the forward computation through an exact-zero graph dependency. These dormant tensors equalize the optimizer and DDP inventory but are excluded from HSTU's active parameter count.

For ML-1M, each seed's metric is averaged over epochs 96–100 before computing the across-seed mean. For ML-20M, both arms use the final full-corpus evaluation at epoch 100. Amazon uses its final full-corpus evaluation at epoch 200; its paired runs are still in progress and are not reported.

Results. Table 2 places our measurements beside the public results reported by HSTU. On ML-1M, SAFA improves every reported metric. Mean NDCG@10 increases by 0.00558, with paired seed deltas of +0.00233, +0.00922, and +0.00520. The positive delta in all three pairs indicates that the aggregate gain is not driven by a single initialization. HR@50 also improves consistently, from 0.59715 to 0.60325, suggesting that forgetting improves the broader candidate ranking in addition to its top positions. On ML-20M, mean NDCG@10 increases from 0.21094 to 0.21512 (+0.00418), with paired deltas of +0.00364, +0.00376, and +0.00513. HR@10 increases from 0.35625 to 0.36228, and HR@50 from 0.61474 to 0.62068; every paired metric delta is positive.

Dataset	Model	Active params.	NDCG@10	HR@10	HR@50
ML-1M	SASRec (2023)*	–	0.1603	0.2853	0.5474
	BERT4Rec*	–	0.1537	0.2843	–
	GRU4Rec*	–	0.1648	0.2811	–
	HSTU*	–	0.1720	0.3097	0.5754
	HSTU-large*	–	0.1893	0.3294	0.5935
	HSTU-large	313,000	0.1902 ±0.0030	0.3318 ±0.0044	0.5972 ±0.0020
	SAFA-large (ours)	313,416	**0.1958 ±0.0012**	**0.3399 ±0.0029**	**0.6033 ±0.0022**
ML-20M	SASRec (2023)*	–	0.1621	0.2906	0.5499
	BERT4Rec*	–	0.1703	0.2816	–
	GRU4Rec*	–	0.1730	0.2813	–
	HSTU*	–	0.1878	0.3252	0.5885
	HSTU-large*	–	0.2106	0.3567	0.6149
	HSTU-large	38,913,120	0.2109 ±0.0002	0.3563 ±0.0003	0.6147 ±0.0006
	SAFA-large (ours)	38,917,344	**0.2151 ±0.0008**	**0.3623 ±0.0007**	**0.6207 ±0.0011**
Amazon Books	SASRec (2023)*	–	0.0156	0.0292	0.0729
	HSTU*	–	0.0219	0.0404	0.0943
	HSTU-large*	–	0.0257	0.0469	0.1066

Table 2: Public sequential recommendation results. Rows marked * are copied from Table 12 of the HSTU paper [Zhai et al., 2024]; they are published point estimates, not our reruns. Our rows report mean ± sample standard deviation over seeds 42–44. ML-1M averages epochs 96–100; both ML-20M arms use the final evaluation at epoch 100. Active parameter counts exclude dormant control gates used only to match the training inventory.

4. Related Work

Generative recommendation and HSTU. Generative Recommenders cast recommendation as causal sequence modeling and introduce HSTU as a unified block for feature interaction and sequential aggregation [Zhai et al., 2024]. HSTU's pointwise SiLU attention avoids row normalization so the amount of supporting history remains observable, and its relative bias incorporates both position and elapsed time. SAFA retains this layer and adds an explicit survival process between each key and query.

Forgetting in attention. The Forgetting Transformer introduces a data-dependent retention process for softmax attention [Lin et al., 2025]. SAFA uses the same general idea of multiplicative path survival but applies it to HSTU's signed, unnormalized coefficients. This distinction determines the construction: survival multiplies outside SiLU so it attenuates magnitude without altering sign or normalizing the remaining events. Learned token-retention methods instead target memory-bounded autoregressive inference: they score tokens and evict cached keys and values under a fixed budget [Bui et al., 2026a,b]. SAFA performs no cache eviction; its survival gate scales each retained key–query pair's contribution inside the sequence mixer.

5. Limitations

Our controlled evidence is limited to three seeds per MovieLens benchmark, and the Amazon paired runs are still in progress. All three public benchmarks are much smaller than industrial recommendation streams. The gate is one scalar per head and event, so it expresses a shared boundary within each head rather than feature-wise survival. Finally, the reported quality experiments use the dense PyTorch reference; we have not yet reported Triton throughput or memory measurements at industrial sequence lengths.

6. Conclusion

SAFA adds one missing operation to HSTU: explicit survival across the path between a historical action and the current query. A learned boundary attenuates every pair that crosses it, while HSTU's signed SiLU coefficients, relative time and position bias, and additive value aggregation remain unchanged. The controlled comparisons improve NDCG@10 in all six paired seeds across MovieLens-1M and MovieLens-20M. These results support path-dependent forgetting as a useful inductive bias for generative recommendation.

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